Fishing for Sustainability — Instructor Setup Sheet

One-Page Facilitation & Tracking

Reproduction Rule: After harvest each round, fish remaining in the lake **double**, up to a **maximum of 30**.

Pre-Class Checklist

- ☐ Bowl with **30** fish (tokens)
- ☐ **8** cups (boats)
- ☐ Extra tokens (10–20)
- ☐ **32** printed student handouts (R1, R2, R3, Debrief)
- ☐ Pens/pencils & scratch paper (8)
- ☐ (Optional) Whiteboard + markers, calculator

Timing Guide (target)

Segment	Time
Round 1 (no discussion)	10 min
Round 2 (quota; 2-min discussion)	12 min
Round 3 (per-person; 2-min discussion)	10 min
Final Debrief handout	8–12 min

Round Trackers (fill during class)

Round 1 — Open Access (no communication)

Field	Value
Start fish	30
Total class catch	
Fish remaining after harvest	
Fish after reproduction (double, max 30)	

Round 2 — Fixed Quota (2-min group discussion to set total quota)

Field	Value
Start fish	30
Agreed class quota	
Total class catch	
Over quota? (Y/N)	
Fish remaining after harvest	
Fish after reproduction (double, max 30)	

Round 3 — Fixed Effort (2-min discussion to set per-person limit)

Field	Value
Start fish	30
Agreed per-person limit	
Total class catch (limit $\times$ 8)	
Fish remaining after harvest	
Fish after reproduction (double, max 30)	

End-of-Class Reminders

- Collect **all four** student sheets per student for participation credit.
- Optional: candy “fish” rewards.
- Jot debrief highlights to seed exam items:

Quick Debrief Points:
- **Tragedy of the Commons:** incentives vs. collective outcomes.
- **MSY & K (30):** Highest sustainable yield occurs when stock left post-harvest is near ~½K.
- **Policy tradeoffs:** Quota (monitoring/enforcement) vs. effort limits (equity/compliance).
- **Transferability:** forests, groundwater, wildlife, carbon.